

HASAN TATAR

Full-Stack Engineer · Applied AI & LLM Systems

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SUMMARY

Software engineer building since 2018, now shipping production Python/FastAPI and TypeScript/Next.js systems for paying clients, with applied-AI work in production. First author on an IEEE SIU 2026 paper on Turkish-language degradation in a vision-language-action model. B.Sc. Computer Engineering expected 2027. Open to full-time, contract, and freelance roles in Istanbul or remote.

EXPERIENCE

Freelance Software & AI Engineer

Jan 2026 – Sep 2026

Independent contractor, Istanbul (remote). Parallel client engagements listed below.

US fleet-operations company · NDA client · bus fleet operations platform

May 2026 – Sep 2026

- Designed a two-plane architecture separating high-frequency location telemetry (HTTP API, Lambda, DynamoDB) from consistency-critical CRUD and live subscriptions (AWS AppSync GraphQL), across 5 CDK stacks and 48 resolvers.
- Encoded regulatory hours-of-service limits as a unit-tested pure module, including multi-driver crew math where a two-driver crew is legal on combined remaining hours neither driver could cover alone.
- Found four classes of silent AppSync resolver failure invisible to TypeScript, lint, and CDK synth by driving the platform's own evaluate-code API, then wired automated validation of all 48 resolvers into CI.

Apparel e-commerce brand · NDA client · AI customer service and ad automation

May 2026 – Sep 2026

- Built the orchestration layer of an async AI customer-service pipeline (FastAPI, async SQLAlchemy, PostgreSQL) ingesting HMAC-verified webhooks from four messaging channels through a ten-stage turn: security filter, intent classification, order lookup, model reply, output validation, send, persist, escalate.
- Cut model spend by short-circuiting intent classification with two zero-token heuristics ahead of any model call; roughly 67% of conversations were classified without one.
- Restructured the model request into cached and per-turn blocks after finding the previously fused system prompt broke provider-side prompt-cache prefix matching on every call.
- Eliminated duplicate customer messages during rolling deploys with a Postgres transaction-scoped advisory lock keyed on conversation id, layered over the existing per-process lock, with a regression test reproducing the race.

GIS Analytics · freelancer network, London, UK · client projects routed through the network

Mar 2026 – Sep 2026

- Built web interfaces with interactive 3D rendering for geospatial data visualization, and full-stack features across the network's product from data layer to client.
- Built an industrial electrical distributor's website and migrated its catalog off a legacy Laravel CMS: an ETL parsing 2,798 bilingual records across 266 categories into 1,875 deduplicated products, with shared canonical and hreflang metadata and a locale-aware sitemap across three locales.
- Provided ongoing technical support to a ready-to-wear fashion brand, including Meta Ads account analysis and a scoped service proposal.

okeep.co / yaptir.io · freelance client · software and automation agency

Jan 2026 – May 2026

- Built the agency's marketing site (Vite, vanilla JS) with a custom three.js WebGL hero and a Groq-backed chat assistant; led a small engineering team and ran client-facing technical discussions across parallel engagements.
- Shipped automation workflows for AI-driven client operations, and closed the engagement with a security pass removing hardcoded credentials from the repository.
- Alongside client work, built Retro Lokal, a vintage storefront on Next.js 16, Supabase, and iyzico with a CraftJS page builder and a Flutter companion app.

Pinsoft, IT Solutions and Consulting · Istanbul

Jun 2025 – Dec 2025

Research & Development Intern Aug 2025 – Dec 2025 · Software Intern Jun 2025 – Aug 2025

- Built an image-based lost-baggage matching pipeline combining color, shape, and texture similarity with multi-angle, centroid-weighted voting to re-identify luggage across camera angles; sole author, shipped across 29 versioned releases.

- Integrated ESP, AD, and ADS hardware modules with the imaging software for global airline companies.

Earlier and part-time roles

- **B4AFC, Business for Aging Friendly Community**, IT Specialist (part-time), Jan–Mar 2026: designed, built, and maintained the organization's website and internal digital tooling.
- **Independent**, Game Developer, Aug–Dec 2024: shipped an isometric RPG and a 2D office game in Unity and C#, handling gameplay design and art direction end to end.
- **NOVENTIQ Makronet Bilgi Teknolojileri**, Software Intern, Jul–Aug 2024: contributed to internal .NET and web tools; supported the data analytics team on reporting workflows.

SELECTED PROJECTS

Quantum: Windows-on-Apple-Silicon compatibility layer *Rust · github.com/wazder/quantum*

Runs an unmodified Windows D3D11 binary end to end on Apple Silicon. A translation stack written from scratch with no third-party crates: PE/COFF loader, x86-64 decoder, AArch64 emitter validated against clang-assembled golden output, a register-pinned JIT lifter, and a DXBC shader parser emitting Metal Shading Language into a D3D11-to-Metal device layer. Seven-crate Cargo workspace, 84 tests, clippy-clean under -D warnings.

CamelLayer: prompt optimization API for non-English AI coding *Python, TypeScript · [live product · camellayer.com](https://liveproduct.camellayer.com)*

Detects the language of a developer's prompt, fixes typos, compresses it, and translates it to English before it reaches the model, wired into Claude, Gemini, Codex, and Copilot CLIs through hooks, a shell wrapper, and a proxy mode. Benchmarked on 1,600 test cases across 14 categories: 100% pass rate, 41.3% average token savings, zero false positives on English input, and code blocks preserved verbatim. Free and Pro tiers with per-key rate limits, served from Google Cloud Run.

NEST: open-vocabulary EEG-to-text decoding *Python, PyTorch · github.com/wazder/NEST*

Neural EEG Sequence Transducer: a Transformer encoder over word-level frequency-band EEG features (105 channels across 8 bands) from the ZuCo corpus, cross-attended by a partially frozen BART decoder, with a CTC baseline. Subject-independent protocol holding out two readers never seen in training, free autoregressive decoding without teacher forcing, and a noise-input control following Jo et al. (2024). Training and evaluation pipeline complete; the full-scale run is pending GPU access.

PUBLICATION

H. Tatar, U. Gökmen, I. Akkoyun, and S. Kahraman, “Performance Analysis of Vision-Language-Action Models: Evaluating Turkish Command Understanding in OpenVLA,” in *Proc. 2026 34th Signal Processing and Communications Applications Conf. (SIU)*, İstanbul, Türkiye, Jul. 2026, pp. 1–4, doi: [10.1109/SIU71813.2026.11636432](https://doi.org/10.1109/SIU71813.2026.11636432).

First and corresponding author. 50 manipulation commands in matched Turkish/English pairs over 300 OpenVLA-7B inferences: mean L2 action error 0.501, cosine similarity 0.372 ($p < 0.001$), complexity uncorrelated to degradation, and 30.8% more tokens for Turkish after BPE segmentation.

EDUCATION

İstanbul Ticaret Üniversitesi · B.Sc. Computer Engineering	Sep 2023 – Expected 2027
Fatih Sultan Mehmet Vakıf Üniversitesi · B.Sc. Computer Engineering, transferred	Sep 2022 – Jul 2023

SKILLS & CERTIFICATIONS

LANGUAGES	Python · TypeScript · JavaScript · Rust · Java · C# · C · Swift · SQL
BACKEND	FastAPI · Node.js · Spring Boot · .NET · PostgreSQL · SQLAlchemy · Supabase · REST APIs
FRONTEND	React · Next.js · Astro · Tailwind CSS · three.js
AI	PyTorch · Transformers · Anthropic Claude API · Computer Vision (YOLO/SAM inference and tracking)
INFRA	AWS CDK · Google Cloud · Cloudflare Workers · Vercel · Firebase · Docker · Git · CI/CD
SPOKEN	Turkish (native) · English (professional)
CERTIFICATIONS	Google Cloud Professional Cloud Architect · Anthropic Claude Developer track (Claude Code, Claude API, Agent Skills, MCP) · Yapay Zeka ve Teknoloji Akademisi Advanced AI program